

Fig. 6: Wiring port definition and terminal wiring sequence of the sensor

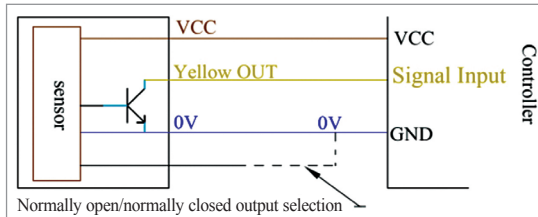


Fig. 7: Normally open or normally closed output selection

Construction and testing

Assemble the circuit as per the circuit diagram shown in Fig. 3. Refer Fig. 6 and Fig. 7 before assembling the project. Fig. 6 shows wiring port definition and terminal

wiring sequence of the sensor while Fig. 7 shows how to select normally open or normally closed output. The value of the threshold in the sensor can be changed with the help of the sensitivity screw, which is shown in Fig. 9 with an opaque container having greater wall thickness as compared to the transparent container.

Product Applications: [NPN output drives small electric relays (coil current $\leq 100\text{mA}$)]

1. When the black line is sus-

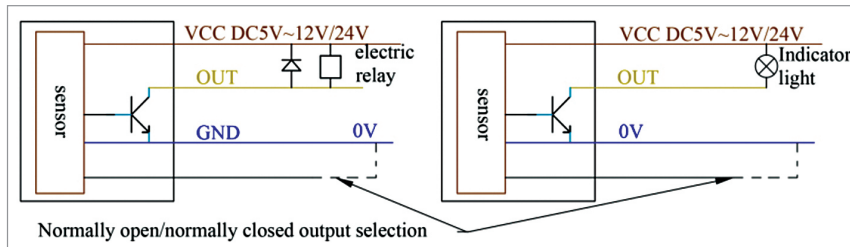


Fig. 8: Connection of an electric relay or an indicator light for interfacing

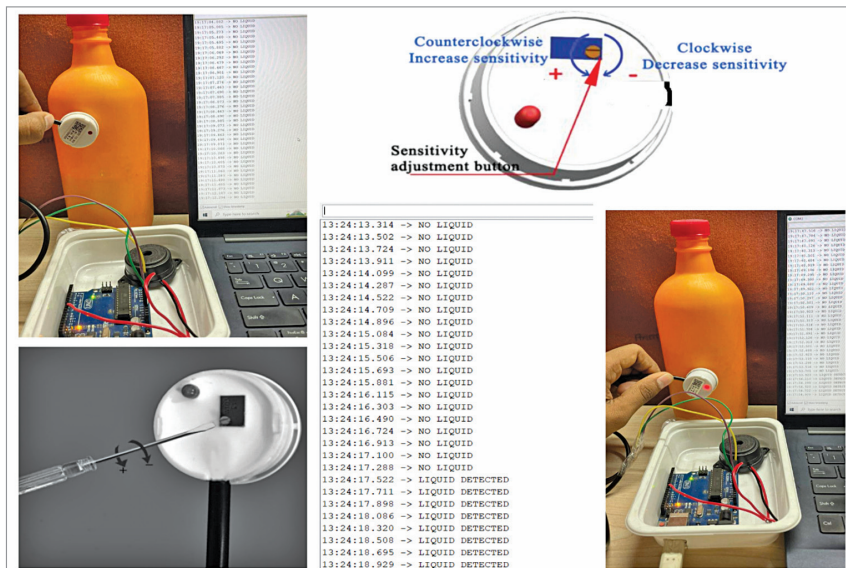


Fig. 9: Operation of the sensor using opaque containers

ended, it is normally open output. If liquid is sensed, the transistor is turned on and closed and the relay is energised and closed. When no liquid is sensed, the transistor is cut off and the relay does not close when the power is turned off.

2. When the black wire is grounded (connected to the negative pole of the power supply 0V), it is a normally closed output. If liquid is sensed, the transistor is cut off and the relay does not close when the power is turned off. When no liquid is sensed, the transistor is turned on and closed and the relay is energised and closed.

Fig. 8 shows the connections for both normally open and normally closed conditions for an electric relay or a light indicator. Other variants of this same sensor work at higher voltages.

Fig. 9 shows the working of the proposed setup for opaque containers (with greater material thickness) by varying the sensitivity of the sensor, which actually makes this sensor more attractive as it is a totally non-invasive approach. The serial-monitor prints LIQUID DETECTED as soon as the sensor detects the liquid and the buzzer makes a beeping sound. The output of the sensor is read by the Arduino on analogue-read pin and it initiates buzzer conditional action accordingly.

Before assembly do not forget to upload the code `liqleveldetect.ino` to Arduino by selecting the port and board name. Then, on opening the Arduino serial monitor, the reading of liquid detection can be seen in serial monitor. When the liquid level rises to the sensor the buzzer sounds and the data is displayed on serial monitor. **EFY**

Dr Geetali Saha is a faculty in the Department of Electronics and Communication, G.H. Patel College of Engineering and Technology (GCET). Under her guidance, Yugmil Patel, final year MC student has implemented the code.

